Curriculum Vitae

Liangliang Xiang

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Research Interests

Gait; Motor control; Rehabilitation; Wearable sensing; Biomechanics; Lower limb modelling; Human performance enhancement; Foot-ankle modelling; Sports Injuries, etc.

Research Skills

Gait analysis; Motion capture; Wearable sensors; Musculoskeletal modelling; Medical image analysis; Finite element modeling (Abaqus, Ansys, FEBio); Statistical Shape Modelling; Machine learning; Reinforcement learning

Research Summary

Postdoctoral Researcher at KTH Royal Institute of Technology with extensive expertise in **wearable-AI** and **gait biomechanics**. Currently serving as **Principal Investigator (PI)** for over **2.6 million SEK** in research grants focusing on digital twins and precision rehabilitation. Author of **40+ SCI publications** (15+ as first/corresponding author, including **ESI Highly Cited** works). Dedicated to translating complex biomechanical data into actionable insights for human performance and clinical rehabilitation.

Open-Source Software & Simulations

- **Deep Learning Framework for Bone Stress Prediction:** Developed a domain adaptation-based LSTM framework for real-time biomechanical inference. Source code available at: https://github.com/Biomechicshub/Bone_stressPred
- **Statistical Shape Models of the Foot:** Hosted open-access datasets of human foot statistical shape models. Data available at: https://doi.org/10.5281/zenodo.13297928
- **Deep Reinforcement Learning for Muscle Redundancy:** Provided preliminary demonstration and results of a DRL framework for real-time muscle state inference. Demo available at: https://github.com/Biomechicshub/DRL-with-MSK-for-MRS

Education

Sept. 2020 – Nov. 2024	University of Auckland
Auckland, New Zealand	Major：Bioengineering
	Ph. D.
Sept. 2017 – Jun. 2020	Ningbo University
Ningbo, China	Major：Biomechanics
	M. Ed.
Sept. 2016 – Dec. 2016	Guangzhou University
Guangzhou, China	Visiting Student
Sept. 2013 – Jun. 2017	Qiannan Normal University of Nationalities
Duyun, China	Major：Kinesiology
	B. Ed.

Research Grants & Funding

- **2025–2027: Digital Futures Research Grant, RISE Research Institutes of Sweden**
  - **Project**: Data-Driven Gait Biomechanics for Precision Rehabilitation
  - **Role**: Principal Investigator
  - **Amount**: 2,000,000 SEK
- **2026–2027: Swedish Research Council for Sports Science (CIF) Grant**
  - **Project**: Tibia Digital Twin for Runners: From Smartphone Scans to Wearable-AI prediction
  - **Role**: Principal Investigator
  - **Amount**: 614,000 SEK

Teaching Experience

Teaching Assistant for Solid Mechanics at KTH Royal Institute of Technology, Sweden (September 2025 – January 2026)

Conference Papers

1. The effects of foot pronation during distance running on the lower limb impact acceleration and running stability. ***Oral presentation***, July 10-14, 2022, Taiwan. (*The 9th World Congress of Biomechanics*)

2. A LSTM framework for ankle joint biomechanics predictions from inertial sensors during gait. ***Oral presentation***, July 9-12, 2023, Maastricht, Netherlands. (*The 28th Congress of European Society of Biomechanics*)
3. Personalized 3D foot-ankle modelling using statistical morphometry analysis. ***Oral presentation***, July 30 - August 3, 2023, Fukuoka, Japan. (*The 29th Congress of the International Society of Biomechanics*)
4. Integrating personalized models, wearables, and deep learning for predicting foot bone stress during running. ***Oral presentation***, June 30-July 3, 2024, Edinburgh, Scotland. (*The 29th Congress of European Society of Biomechanics*)
5. A data-driven pipeline for smartphone-based photogrammetric 3D foot reconstruction. ***Oral presentation***, July 27-31, 2025, Stockholm, Sweden. (*The 30th Congress of the International Society of Biomechanics*)
6. From smart scanning to AI-enabled wearables: A digital health pipeline in foot biomechanics ***Oral presentation***, October 13-15, 2025, Budapest, Hungary (*3rd Biomechanics in Sport and Ageing Symposium*)
7. A digital twin framework for personalized stroke gait rehabilitation using neuromusculoskeletal modeling and deep reinforcement learning. ***Poster presentation***, November 11, 2025 (*Digitalize in Sthlm'25*)

Selected Publications

1. **Xiang L.,** Gu Y., Deng K., Gao Z., Shim V., Wang A. & Fernandez J. (2025). Integrating personalized shape prediction, biomechanical modeling, and wearables for bone stress prediction in runners. *npj Digital Medicine,* 8, 276.
2. **Xiang L.**, Gao, Z., Fernandez, J., Gu, Y., Wang, R., & Gutierrez-Farewik, E. M. (2025). Explainable artificial intelligence for gait analysis: Advances, pitfalls, and challenges - A systematic review. *Frontiers in Bioengineering and Biotechnology*, 13, 1671344.
3. **Xiang L.,** Gu Y., Gao Z., Yu P., Shim V., Wang A. & Fernandez J. (2024). Integrating an LSTM framework for predicting ankle joint biomechanics during gait using inertial sensors. *Computers in Biology and Medicine,* 170, 108016.
4. **Xiang L.**, Gu Y., Shim V., Yeung T., Wang A. & Fernandez J. (2024). A hybrid statistical morphometry free-form deformation approach to 3D personalized foot-ankle models. *Journal of Biomechanics*, 168, 112120.
5. **Xiang L.**, Gao Z., Wang A., Shim V., Fekete G., Gu Y. & Fernandez, J. (2024). Rethinking running biomechanics: A critical review of ground reaction forces, tibial bone loading, and the role of wearable sensors. *Frontiers in Bioengineering and Biotechnology*, *12*, 1377383.

6. **Xiang L.**, Gu Y., Wang A., Shim V., Gao Z. & Fernandez J. (2023). Foot pronation prediction with inertial sensors during running: A preliminary application of data-driven approaches. *Journal of Human Kinetics*, 88, 163059.
7. **Xiang L.**, Mei Q., Wang A., Shim V., Fernandez J. & Gu Y. (2022). Evaluating function in the hallux valgus foot following a 12-week minimalist footwear intervention: A pilot computational analysis. *Journal of Biomechanics*, 132, 110941.
8. **Xiang L.**, Gu Y., Mei Q., Wang A., Shim V. & Fernandez J. (2022). Automatic classification of barefoot and shod populations based on the foot metrics and plantar pressure patterns. *Frontiers in Bioengineering and Biotechnology*, 10, 843204.
9. **Xiang L.**, Mei Q., Fernandez J. & Gu Y. (2020). A biomechanical assessment of the acute hallux abduction manipulation intervention. *Gait & posture*, 76, 210-217.
10. **Xiang L.**, Mei Q., Fernandez J. & Gu Y. (2018). Minimalist shoes running intervention can alter the plantar loading distribution and deformation of hallux valgus: A pilot study. *Gait & posture*, 65, 65-71.

（A full list of 40+ publications is available upon request or via Google Scholar）

Awards & Honors

- **Distinguished Undergraduate Graduate of Guizhou Province, China (2017)**
- **Distinguished Master's Graduate of Zhejiang Province, China (2020)**
- **2023 Ningbo Municipal Science and Technology Progress Award, Runner-up.** Biomechanical research on the movement function of the metatarsophalangeal joint and innovation in footwear equipment. (5/8)
- **2024 Top Cited Articles of the Year, Journal of Biomechanics** (#3)
- **2024 Digital Futures Research Fellowship, supported by the Research Institutes of Sweden**, KTH Royal Institute of Technology, Sweden

Academic Service

- **Editorial Board Member & Editor:** *BMC Musculoskeletal Disorders* (2023–) , *BMC Sports Science, Medicine, and Rehabilitation* (2023–) , *Frontiers in Sports and Active Living* (2022–) , and Assistant Associate Editor for *Computer Methods in Biomechanics and Biomedical Eng.* (2022).
- **Invited Reviewer:** *BMC Medicine*, *Sports Medicine*, *IEEE JBHI*, Journal of Biomechanics, Journal of NeuroEngineering and Rehabilitation, *Gait & Posture*, among others.